

Proforma Accuracy and Real Revenue Analysis in Furnished Housing Solutions

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Abstract

What drives revenue to deviate from the economic models used at CorDuro? Which variables matter the most for the stabilization of the portfolio? Using rent roll data and the Proformas generated by the salespeople, this paper measures the impact different factors have on real revenue variance.

1 Introduction

CorDuro (CD) is a furnished housing solution provider that sells their product both ways: to Members and Partners...

In September 2025, the Investment Committee was established to...

One of the main methods to verify investment opportunities is using an economic model called a Proforma. Each of these is tied to a specific Deal, not Property. A Property can have multiple Deals, each corresponding to a Proforma. This document sources its data from... it contains data... NOI...

The Proforma is then generated by each salesperson using the information available on the Partner's website at the time of creation. Each salesperson must calculate the average rent cost for each of the unit types that CD is interested in acquiring (Studios, 1-Bedroom units, 2-Bedroom units, and 3-Bedroom units) at that Property using the listed rent for the units at the Property's website. After adding these values, the Proforma calculates a blended rent by weighing the average rents per unit type (weighted average). This Proforma is then used by the salesperson as a sales tool to get a potential Partner on board with CD's program.

After, the pitch moves into the creation of an Order Form... Partner signs... the order would have CD... acquire keys... operation's team... paint... furnish... ready for launch and be rented out to our Members.

It is before the field operations occur where the first opportunity for discrepancy arises. Partners are not contractually obligated to deliver the mix of units that the salesperson used to create the Proforma. Furthermore, Partners are not obligated to provide units that average the blended rent that the Investment Committee approved. After a Deal is signed, the Partner fills out a form, the Info Sheet, with the details of the units they'll provide to CD.

Insert analysis of distribution for Proforma vs Info Sheet unit mix.

This variation between the projected revenues from the Proforma and the Info Sheet is the first step into understanding how robust the Proformas truly are. However, the goal is not to compare a projection to another projection. The objective of this paper is to understand the reasons why, beyond this simple comparison, real revenue deviates from the projected revenue of the Proformas. The reasons being that...

So, why does real revenue deviate from the projected revenue?

2 Data and Research Design

To answer the research question, data will be collected all throughout 2026, starting with the deals that were signed on January of the same year. To avoid prorating rent values on the Proforma, we will exclude the first month of launch ($t = 0$) from the regression entirely unless the deal goes live on the 1st of the month. We cannot assume that revenue is linear. However, all information will still be gathered for all variables. The information gathered includes all the original Proformas for these deals, as well as the real revenue generated by the corresponding units using monthly released information on Rent Roll. Other sources are... the information provided by the Proformas had to be calculated further to obtain the revenue CD would keep. This has been constructed with our books, which include the detailed reservation history per property...

The information has been arranged into a panel data set (longitudinal data) in long format where every deal is being tracked across multiple time periods. Each period has its own row, as required by standard regression software. For this paper, data is collected and arranged on an Excel sheet and exported to RStudio for analysis. Since there can be multiple deals per property, rent roll is available on a monthly basis, and

the Proformas project revenue for each month since the deal goes live, the unit of analysis is defined as deal-month (i, t) .

Since deals may incorporate any number of units from five, twenty, to fifty, and their blended rates vary from one market to the next, nominal numbers cannot be used as a comparison. To ensure statistical comparability across deals of varying unit counts and market rent values, this paper will use a percentage ratio between CD's Proforma projected revenue and CD's real revenue. The selected dependent variable is the *revenue performance variance*, such that:

$$Y = \frac{\textit{Actual Net Rent} - \textit{Proforma Net Rent}}{\textit{Proforma Net Rent}}$$

If Y is positive, CD's real revenue has exceeded expectations. If Y is zero, it indicates that the Proformas are accurately predicting CD's real revenue. If Y is less than zero, then the models are underperforming and CD's real revenue falls short of expectations.

3 Methodology: The Determinants of Variance

To explain the logic of how real revenue behaves, we will use a mix of variables that attempt to control through structural factors, which test for rental condition and quality assumptions; market dynamics, which quantify external risks; and operational execution, which measure efficiency. Additionally, we account for fixed effects by addressing the human factor in the creation of a Proforma. Since the data being used is longitudinal, where months one, two and three of a deal are not strictly independent from each other, we will cluster standard errors at the deal level for a robustness check. This correction should prevent false positives caused by correlation from showing up.

3.1 Theoretical Framework and Hypothesis

Introduction.

3.1.1 Structural Variables

We propose that existing units reduce the marginal cost of acquiring new units at the property and shorten the ramp-up period because the building is already stabilized in marketing channels. **H1: preexisting presence in a property drives down the ramp-up period, positively increasing revenue variance.** To test this, we employ

the variable **Expansion Ratio** (X_{Exp}), calculated as a continuous ratio between the existing units and the new units.

We theorize that Member’s demand for furnished housing is structurally concentrated in specific unit types (1-Bedroom and 2-Bedroom units), making deviations from this “core” mix riskier. Assets with a high concentration of studios or large multi-bedroom units face lower demand, leading to higher vacancies and lower revenue. **H2: Properties with a higher ratio of non-core units will exhibit lower revenue performance.** To test this, we employ the variable **Non-Core Unit Ratio** (X_{Mix}), calculated as the proportion of Studios and 3-Bedroom units relative to the total unit count.

Endogeneity Bias. If the Partner delivers a unit mix with a lower blended rent, variance is expected to change from the start. A big variance in revenue could simply be explained by the fact that the unit mix the Partner gave us wasn’t as valuable as the salesperson thought. To account for this, we will use **Initial Blended Rent Deviation** ($X_{InitDev}$), which is the percentage variance between the Proforma and the Info Sheet blended rate.

Although square footage alone does not act as a proper predictor for revenue, since we have a standard, brand-friendly size, it does need to be added as a check

against $\mathbf{X}_{Gap}$. By considering size, we account for omitted biases in other variables.

This variable is defined as **Unit Size** ($\mathbf{X}_{SqFt}$).

Revenue in furnished housing is driven by the tradeoff between occupancy stability and daily rate yield. We propose that short-term rentals (STR), despite higher turnover costs, generate revenue that outweighs the stability of long-term rentals (LTR). So far, we do not have enough information to test this. Every bit of data available shows that long-term rentals produce higher yields due to how high turnover costs are. We expect that, in the near future, these costs will be more efficient. **H3: A higher composition of short-term bookings will result in higher revenue performance variances, up until turnover costs create a plateau.** This hypothesis is tested using **Lease Term Structure** ($\mathbf{X}_{Term}$), calculated as the ratio of days occupied by STR members (stays under 30 nights) between total occupied days.

Online Travel Agency (OTA) reservations carry high channel fees, which directly depress revenue. STR and OTA are not correlational. As of January 2026, four fifths of all STR reservations come from OTA channels, but this number is dwindling due to efforts to transfer OTA reservations into CD's website. Within a year, we expect STR reservations to come mainly from our own channels. For that reason, we will consider the ratio of reservations that come from OTA channels using variable $\mathbf{X}_{OTA}$. **H4: The**

positive revenue effect of STR (X_{Term}) is minimized by OTA dependence. To

test this, we employ the interaction term:

$$X_{Term} \times X_{OTA}$$

Revenue share is not fixed. It can be renegotiated to be higher or lower, which directly impacts revenue variance from the Proforma. Changes in revenue share are exogenous to the operations and markets, so it must be isolated. To control for this, we add a continuous variable, **Revenue Share Difference ($X_{RevShare}$)** that tracks the difference between the revenue share each month and the original revenue share used in the Proforma.

$$X_{RevShare} = AMG\ Revenue\ Share_{it} - AMG\ Proforma\ Revenue\ Share_{it}$$

3.2.1 Market variables

Launch timing dictates the initial trajectory of a deal's revenue. Properties launched during low-demand windows have difficulties recovering in the short term. **H5:**

Launching during a low demand season will have a negative impact on early revenue performance. We operationalize this with a dummy variable **Seasonality**

of Launch (X_{season}), where 1 indicates a high-season launch and 0 a low-season launch, based on the Investment Committee’s definitions. [I am aware that this is too simplistic. Off season is more complex than a dichotomous variable. I need Finances input for this.]

We assume that high-tier markets provide a floor for revenue performance even during low-season windows. Conversely, secondary or tertiary markets lack this floor.

H6: Properties in higher-tier market classes will exhibit a positive correlation with revenue increase. To measure this, we use **Market Class (X_{class})**, an ordinal variable stratifying properties based on the historical strength and presence of CD in the local market.

Members in furnished housing often prioritize convenience over price, making assets in walkable, transit-oriented areas less sensitive to low-demand windows. We argue that location serves as a proxy for demand inelasticity. **H7: Higher location desirability scores correlate with positive revenue performance.** This is operationalized using **Walkable Score (X_{walk})**, derived from the property’s Walk Score (0-100), a standard metric for walkability and transit access.

We propose that the price premium for furnished housing is elastic only up to a particular threshold. As the gap between the marked-up furnished apartments and the surrounding unfurnished apartments widens, the value proposition for Members

diminishes, increasing the likelihood of negative revenue variance. **H8: Premiums over the external market increase revenue up to a threshold, where higher premiums will negatively impact revenue performance.** We test this using **Percentage Premium Over Market (X_{Gap})**, a variable measuring the percentage difference between the property's actual blended rent and a mirror mix external market rent derived from third party data sources.

3.3.1 Operational Execution

The speed at which a property reaches stabilized status (set at an occupancy rate of 90%) is the primary indicator of operational efficiency. Deviations from the projected ramp-up timeframe represent a direct loss of potential revenue that cannot be recaptured and have an impact on the rest of the deal's term. **H9: Delays in stabilization relative to the Proforma will negatively impact revenue performance.** This is measured by **Stabilization Lag (X_{stab})**, calculated as the difference between the projected Proforma ramp months and the actual months taken to reach stabilization.

Higher maintenance volume is not only an operational expense, but a sign of underlying asset quality issues that lead to Member dissatisfaction. We argue that high maintenance requests acts as a drag on net revenue. **H10: Higher maintenance costs will be associated with lower revenue performance.** We operationalize this with **Maintenance Intensity (X_{Maint})**, which is the aggregated cost of maintenance tickets across all units for the observed deal and month.

Gross Revenue in the Proforma includes fixed utility costs, but fluctuations in weather and consumption patterns can create artificial variance that does not reflect true rental performance. **H11: An increase in utility revenue collection will correlate with higher revenue variance, independent of rent performance.**

We control for this using **Utility Deviation** ($X_{UtilDev}$), which is the ratio of the difference between actual and Proforma utility revenue relative to total Proforma revenue.

As a sales tool, we will sometimes offer to waive DURASTEEL Fees for three months, driving down our revenue. This is simply accounted for by a dummy variable X_{Waiver} indicating if there were any DURASTEEL Fee discounts in the unit of analysis (deal-month).

3.4.1 Fixed Effects

Finally, we take into account that Proforma accuracy is a function of not just market forces, but of the salespeople's work. Some salespeople may be systematically optimistic or pessimistic during their underwriting assumptions. **H12: Individual salespeople will exhibit systematic, non-zero biases in their revenue projections.** We isolate this human error using **The Underwriter Effect** (X_{Sales}), a vector of fixed-effect dummy variables for each salesperson, with one of them serving as the baseline intercept.

3.2 Data and Operationalization

H1: preexisting presence in a property drives down the ramp-up period, positively increasing revenue variance. To test H1, we employ the variable X_{Exp} , calculated as a continuous ratio between the existing units and the new units.

$$X_{Exp} = \frac{N_{Exist}}{(N_{Exist} + N_{New})}$$

Where 0 = a New Market, and the higher the ratio, the smaller the expansion compared to the preexisting units.

Additionally, since we cannot assume that the rate of stabilization is identical for new and expansion deals. We propose that *H13: expansion deals stabilize faster.* For that, we will add an interaction between both variables.

$$X_{Exp} \times X_{Stab}$$

H2: Properties with a higher ratio of non-core units will exhibit lower revenue performance. To test H2, we construct the variable X_{Mix} , calculated as the proportion of Studios and 3-Bedroom units relative to the total unit count.

$$X_{Mix} = \frac{\Sigma(Studios) + \Sigma(3BR+)}{Total Units}$$

H3: A higher composition of short-term bookings will result in higher revenue performance variances, up until turnover costs create a plateau. For H3, we made the variable X_{Term} , which distinguishes the ratio of STR from LTR reservations.

$$X_{Term} = \frac{Total Days Occupied by STR Members}{Total Days Occupied by All Members}$$

We expect this variable to perform like a logarithmic curve, where the presence of STR creates a quick rise in revenue, until high turnover and maintenance costs flatten the curve.

H5: Launching during a low demand season will have a negative impact on early revenue performance. For H5, we designate a dummy variable X_{Season} that indicates whether

the deal was launched during a high demand season, defined by information provided by the Investment Committee. A 0 indicates that the deal was launched during low season, a 1 indicates otherwise.

H6: Properties in higher-tier market classes will exhibit a positive correlation with revenue increase. To test H6, we employ X_{class} , a categorical, ordinal variable that stratifies properties according to the historical strength and presence of CD within the local market. The tiers are provided by information of the Investment Committee.

H7: Higher location desirability scores correlate with positive revenue performance. For H7, we use X_{walk} , which is the industry-standard Walk Score (0-100).

H8: Higher rent premiums over the external market will negatively impact revenue performance. To test H8, we construct a variable X_{gap} . This is calculated as the price difference between the deal's actual blended rent and the external market rent (unfurnished), derived through a mirror mix formula from third party sources. The external market rent is calculated as:

$$External\ Rent_{it} = \frac{\sum(R_{Type} \times N_{Type})}{N_{Total}}$$

Where R is the external average rent for a specific unit type and N is the count of those units in the subject property. The final variable is defined then as:

$$X_{Gap} = \frac{(Actual\ Blended\ Rent_{it} - External\ Rent_{it})}{External\ Rent_{it}}$$

As the mark-up exceeds a critical threshold, the value proposition for Members diminishes, having a negative impact on the dependent variable.

H9: Delays in stabilization relative to the Proforma will negatively impact revenue performance. For H9, we define X_{Stab} as the deviation between the projected ramp-up period and the months required to reach stabilization, where:

$$X_{Stab} = Proforma\ Ramp\ Months - Actual\ Months\ to\ Stabilization$$

A positive value indicates the property stabilized faster than predicted; a negative value indicates a delay.

H10: Higher maintenance costs will be associated with lower revenue performance. For H10, we construct X_{Maint} by aggregating the costs of maintenance tickets across all units for the unit of analysis (deal-month).

H11: An increase in utility revenue collection will correlate with higher revenue variance, independent of rent performance. To test H11, we use $X_{UtilDev}$, which is the deviation between the actual utility revenue and the projected utility revenue, where:

$$X_{UtilDev} = \frac{\text{Actual Utility Revenue} - \text{Proforma Utility Revenue}}{\text{Total Proforma Revenue}}$$

For the DURASTEEL Fee Waiver, X_{Waiver} , a 0 means that DURASTEEL Fees were charged for that unit of analysis, whereas a 1 means DURASTEEL Fees were waived.

H12: Individual salespeople will exhibit systematic, non-zero biases in their revenue projections. To test H12, we designate X_{Sales} , a vector of $n - 1$ dummy variables representing the individual salesperson responsible for the Proforma. This fixed effect

isolates human error from market error, using a specific salesperson (Wells Davis) as the baseline intercept.

4 Empirical Results

The relationship between revenue variance and the explanatory variables is then estimated using the following multivariable linear regression equation:

$$\begin{aligned}
 Y_{it} = & \beta_0 + \sum_{j=1}^{n-1} \gamma_j (\text{Salesperson}_j) + \beta_1 X_{Exp} + \beta_2 X_{Stab} + \\
 & \beta_3 (X_{Exp} \times X_{Stab}) + \beta_4 X_{Mix} + \beta_5 X_{InitDev} + \beta_6 X_{SqFt} + \beta_7 \ln(X_{Term} + 1) + \beta_8 X_{OTA} + \\
 & \beta_9 (\ln(X_{Term} + 1) \times X_{OTA}) + \beta_{10} X_{RevShare} + \beta_{11} X_{Season} + \beta_{12} X_{Class} + \beta_{13} X_{Walk} + \\
 & \beta_{14} X_{Gap} + \beta_{15} X_{Gap}^2 + \beta_{16} X_{Maint} + \beta_{17} X_{UtilDev} + \beta_{18} X_{Waiver} + \varepsilon_{it}
 \end{aligned}$$

Statistical significance.

Magnitude of impact.

Robustness check (cluster-robust standard errors). Since the unit of analysis is deal-month (i, t), each deal repeats over time. This is a longitudinal dataset, which will most likely have autocorrelation (observations for the same deal are not independent). To address this problem, standard errors are clustered at the deal level (i). This correlates the errors for each deal over time, while assuming independence across different deals. Failure to cluster standard errors would artificially deflate the

standard errors, increasing the risk of Type I errors (false positives) by treating repeated monthly observations of the same deal as independent data points.

5 Discussion and Strategic Implications

Are adjustments needed?

6 Conclusion

Written last.